PRE-STUDY

Lönn AI

Cost, quality and trust in LLM routing

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Source code: https://github.com/emrailklmin/Lonn

Executive summary

Companies are spending rapidly more on LLMs built into their products through APIs: of an estimated 37 billion USD spent on enterprise generative AI in 2025, 12.5 billion USD went to foundation-model APIs (Menlo Ventures, 2025). This creates a twofold problem. Not every request needs the same capability, so a single model either wastes money on routine tasks or compromises demanding ones. And the total bill does not show which features or workflows generated the spending.

Lönn AI addresses both as a single API layer between a company's applications and its LLM providers. For each request it identifies the type of task and sends it to the model with the best balance of quality and cost, within the company's data-handling requirements. A single parameter, λ, sets how much quality to trade for lower cost, and the same layer is designed to connect spending to the features and teams behind it.

The pre-study covers LLM use through APIs only. It combined a literature and market review, participatory design interviews, a financial model and a prototype of the core routing mechanism, tested on 690 benchmark prompts answered by eight models. Cost tracking, billing and the data-handling safeguards were designed but not implemented.

At its default setting (λ = 20), the Lönn Router answered 85.8% of prompts correctly at 1.58 USD, against 88.1% at 7.58 USD for always using Claude Sonnet 5.5: about 97% of the accuracy at a fifth of the cost. At λ = 10 it reached 89.6% at about half the cost. On everyday prompts, however, the classifier correctly identified the task type for only 51% of prompts, so benchmark results are likely optimistic.

With a fee of 20% of the savings, a customer moving from Claude Sonnet 5.5 still cuts AI spend by about 63%, and Lönn AI covers its non-personnel operating costs at roughly two customers. The cost and quality requirement (R4) is partly met on the benchmark, the reliability requirement (R5) is not yet met, and the business and trust requirements are partly met. The pre-study concludes that Lönn AI is feasible and commercially promising, and recommends pilots with partner companies, first in shadow mode, to validate savings on real traffic.

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1 Definitions and terminology

Term	Meaning in this report
Accuracy (pass rate)	The share of requests a model answers correctly. In the routing table it is measured per task type and model on the benchmark.
Answer check	A check of a model's answer before it is returned. It flags answers that are empty, cut off at the token limit, or contain code that fails the examples in the prompt. A flagged answer can trigger one extra call to a stronger model (escalation).
API (application programming interface)	The interface through which software sends a request to a model provider and receives an answer. Use is billed per token (see Token).
Benchmark	A fixed set of prompts with known correct answers or grading rules. It is used to compare models and routing strategies on the same requests.
Benchmark oracle	A theoretical router that knows every model's result in advance and picks the cheapest model that answers correctly, or the cheapest model when none does. It is an upper bound for what routing can achieve and cannot be built in practice. In the rest of the report it is called the oracle.
Classifier	The model that estimates which task type a prompt belongs to. In the prototype it is a text embedding (bge-small) followed by a linear layer.
Contribution margin	Fee revenue minus variable costs, expressed per routed USD and as a share of fee revenue.
Cross-validation	A way to evaluate on data that was not used to build the router. The prompts are split into folds, and each fold is tested with a routing table built from the others.
Embedding	A numerical vector representation of a text, where texts with similar meaning get similar vectors.
Escalation	Sending a request again to a stronger model after the answer check has flagged the first answer.
Fallback	Sending a request to the next model in the ranking when a provider call fails.
KV cache (key–value cache)	Stored intermediate results that let a provider reuse context that has already been processed. This makes repeated context cheaper and faster. The benefit can be lost when a request switches model.
λ (lambda)	The cost weight in the routing score (Section 3.4). It states how much accuracy one USD of cost per request is worth. Higher values favour cheaper models. The default is $\lambda = 20$.
LLM judge	A language model that grades another model's answer against a set of criteria, used where there is no single correct answer.
LLM routing	Choosing, for each request, which model and provider should handle it.
Middleware	Software that sits between the customer's application and the model providers and forwards requests.
Model pool	The set of candidate models the router can choose between.
Model provider	A company that offers access to a model through an API, for example OpenAI and Anthropic.

Term	Meaning in this report
Pareto frontier	The set of options for which no other option is both cheaper and more accurate.
Reasoning tokens	Tokens a model generates while it "thinks" before answering. They are billed as output even though they are often not shown.
SDK (software development kit)	A code library that makes it easier for developers to call a service from their own programs.
Task type	One of the 15 finer categories the classifier uses. A prompt's task type is the category that the embedding and the classifier together assign to it.
Token	A short piece of text, often part of a word, that models read and write. Usage is billed per token, with output usually priced higher than input.
ZDR (zero data retention)	A setting under which the prompt and the answer are not stored after the request has been processed.

2 Introduction

2.1 Background

Artificial intelligence is becoming part of everyday life, changing how people find information, create content and complete tasks. Organisations are also expanding their use of AI and spending more on it. Menlo Ventures (2025) estimates that enterprise generative AI spending rose from 11.5 billion USD in 2024 to 37 billion USD in 2025.

Organisations use AI both through ready-made applications, such as ChatGPT, and by integrating large language models (LLMs) into their own products and internal systems. In the latter case, software communicates with a model provider through an application programming interface (API), sending requests to a hosted model and receiving responses for use within its own features and workflows (OpenAI, n.d.-a). These integrations account for substantial expenditure: of the estimated 37 billion USD spent on enterprise generative AI in 2025, 12.5 billion USD went to foundation model APIs (Menlo Ventures, 2025).

Providers commonly charge for text-based API use by the token, a small unit of text. Input tokens represent content sent to the model, while output tokens represent content it generates. These are priced separately, with output typically costing more than input. Total spending therefore depends on how many requests are sent, how many tokens each request uses, and which models handle them (Anthropic, n.d.-a).

Choosing a model requires evidence of its suitability for the intended work. Public benchmarks compare models on defined tasks using specified scoring methods, assessing capabilities such as knowledge and reasoning. However, results depend on what is tested and how evaluations are conducted, so strong benchmark performance does not guarantee suitability for every application (Anthropic, 2023). Organizations therefore consider benchmarks alongside internal testing, cost, reliability, security and existing cloud relationships (Andreessen Horowitz, 2025).

These decisions take place in an evolving market, as providers introduce models with different capabilities and price points. Differences in suitability across use cases are already driving multi-model adoption. In a 2025 survey of 100 CIOs across 15 industries, 37% of respondents reported using five or more models, compared with 29% in the 2024 survey. The accompanying buyer research identified model differentiation by use case as the main reason enterprises purchased models from multiple vendors. However, the prompts, guardrails and evaluation processes that an organisation builds are often tuned to one specific model, and may not work as well with another. These dependencies can create *model lock-in*, making switching a matter of engineering and revalidation rather than simply changing an API connection (Andreessen Horowitz, 2025).

Understanding expenditure across these applications requires additional context. Providers expose usage and cost information, but their records do not automatically explain which business activities generated the spending. When applications use several providers, organizations must consolidate information across reporting systems or integrate a shared gateway. Existing tools support this, but workflow-level attribution still requires context, such as identifiers linking requests to teams, features or processes (Anthropic, n.d.-b; LiteLLM, n.d.). The challenge is therefore not simply accessing usage data but making spending consistently traceable across applications and providers while managing the integration and maintenance this requires (FinOps Foundation, 2026).

2.2 Problem

Every AI request has a cost, but not every request needs the same level of capability. Using an expensive model for a routine task can waste money; using a cheaper model for a demanding task can compromise the result. Companies therefore face a cost–quality trade-off: how much is worth paying for the quality a task requires?

Across an application, these decisions accumulate into a broader spending problem. Developers need to deliver reliable performance, while finance needs to understand and control expenditure. Yet knowing the total bill does not explain which features, workflows or customer activities generated it—or whether those costs were justified by the quality delivered. Without that traceability, unnecessary spending is difficult to distinguish from spending that matters.

The problem is twofold: determining the cost needed to meet each task's quality requirements, and tracing expenditure to the activities that make it worthwhile.

2.3 Solution

Lönn AI aims to help companies obtain the quality they need from AI without selecting a model for every request, while providing greater visibility and control over their AI spending.

The solution consists of two connected components: an API-based routing system and a web application. The routing system sits between a company's applications and its AI providers, automatically selecting a suitable model for each request based on estimated quality, cost, and security requirements. The aim is to maintain an appropriate balance as models and prices change, without requiring developers to continually reassess alternatives.

The accompanying web application provides an interactive dashboard where companies can monitor usage and expenditure, analyse routing decisions, compare available models, and configure preferences for quality, cost and security.

The system is also designed to connect spending to the projects, applications and teams responsible for it. This would give developers greater insight into operational costs and help decision-makers understand what drives expenditure.

Together, these components bring model selection, cost optimisation and usage management into one coordinated service.

2.4 Scope

This pre-study has two aims. The first is to analyse theoretically how LLM routing can reduce the cost of using LLMs while maintaining answer quality, and whether this can form the basis of a viable business. The second is to build a prototype that shows the theoretical solution can be realized in practice.

The pre-study is limited to the use of LLMs through APIs. This means that companies send requests to model providers directly from their own products, tools or workflows, and pay per token for each request. The use of AI through subscriptions to chat assistants, such as ChatGPT or Microsoft Copilot, falls outside the scope.

The prototype is limited to the core routing mechanism, meaning identifying the type of task a request belongs to and choosing a suitable model for it. It also sends the request to the selected LLM provider, but only within a limited pool of eight models. It was developed under financial constraints, as generating and evaluating thousands of LLM responses incurs substantial API costs. Furthermore, evaluation was conducted in a controlled setting rather than on real customer traffic, limiting the generalisability of the results to real-world usage.

Other parts of a complete product, such as prompt compression, cost tracking and billing, fall outside the prototype. Data security is treated only theoretically in this pre-study and is not implemented in the prototype.

Future development of these parts is described in the roadmap.

3 Method

3.1 Study design

The project was conducted as an iterative feasibility study of Lönn AI. The study identified the technical, business and trust assumptions on which the proposed service depended and chose an assessment method for each. Routing performance was examined through prototype development and benchmark testing. Customer needs and data-handling requirements were investigated through participatory design interviews and literature review. Commercial feasibility was assessed using a financial model that combined the prototype's measured costs with assumptions about pricing and customer usage.

Findings from one activity informed the next. Stakeholder feedback helped define which routing decisions customers should control, comparisons between prototype versions guided the choice of routing logic, and the measured cost difference between routed and fixed-model requests provided an input to the financial model.

3.2 Literature and market review

The literature review focused on three technical questions: whether models answer different requests correctly, whether a router can accurately predict which model will produce the best output for a task, and whether new models can be added without retraining the routing system. For each paper, the review noted which models and tasks were tested, what the router was compared with, and what cost and quality were reported. Two kinds of results were kept apart. Oracle results show the best outcome that routing could reach with hindsight. Results from implemented routers show what a working system actually achieves.

The market review focused on services that route LLM requests through an API. Competitors were grouped by product focus and deployment model. For the trust assessment, provider documentation on processing location, retention and training use was reviewed to identify which conditions would need to be checked before a customer request is sent to a particular endpoint.

3.3 Stakeholder consultations and participatory design

The study consulted participants with direct experience of three aspects of the proposed service. A founding engineer at Zaplar discussed how LLMs were used across product functions, where model selection could be delegated, and where predictable answer quality was important. This feedback was used to refine the proposed customer controls, including the ability to give cost and quality different weights for different requests.

An operations specialist with experience handling sensitive data advised on retention, hosting, encryption and the documentation a prospective customer might expect. This consultation informed the proposed trust requirements. A separate consultation with a potential customer provided an estimated range of monthly API token usage for a B2B use case. That range was used to construct the customer scenario in the financial analysis.

3.4 Prototype development and benchmark evaluation

Benchmark construction

A common benchmark was constructed to compare routing strategies on the same requests. It comprised 690 prompts across five task families (Table 3.1).

Table 3.1. The benchmark: task families, datasets and number of prompts.

Task family	Datasets	Prompts
Mathematics	MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024)	150
Programming	HumanEval+ and MBPP+ (Liu et al., 2023); LiveCodeBench (Jain et al., 2025)	200
Knowledge	MMLU-Pro (Wang et al., 2024); GPQA Diamond (Rein et al., 2024); ARC-Challenge (Clark et al., 2018)	200
Instruction following	IFEval (Zhou et al., 2023)	80
Business writing	WritingBench (Wu et al., 2025)	60
Total		**690**

Prompts were sampled using a fixed random seed (20261012). MATH-500 and LiveCodeBench were sampled within difficulty strata, MMLU-Pro approximately equally across subjects, and WritingBench equally from two English business-related domains. The remaining datasets were sampled randomly. The number of prompts drawn from each dataset and the sampling details are shown in Table C.1 in Appendix C.

Each of the eight candidate models was run on every prompt. The resulting answers were stored with their grades and recorded costs to form an answer bank for replaying routing decisions. Costs were based on billed model calls, including reasoning tokens where applicable. Grading followed the task: reference answers for questions with a defined solution, executable tests for programming tasks, rule-based checks for instruction following, and an LLM judge applying the WritingBench rubric for business writing.

Table 3.2. The eight candidate models: output list price, share of non-writing prompts answered correctly, and billed cost for all 690 prompts.

Model	Output price (USD per million tokens)	Correct (excl. writing)	Billed cost, 690 prompts (USD)
Claude Sonnet 5.5	10.00	92%	7.58
Gemini 3.8 Flash	3.75	90%	4.52
Muse Spark 1.3	4.25	88%	7.37
DeepSeek V4.1 Flash	0.60	84%	1.16
gpt-oss-120b	0.17	82%	0.29
DeepSeek V4 Pro	0.89	81%	3.98
gpt-oss-20b	0.09	77%	0.18
GLM 5.3 Flash	1.20	75%	0.37

Routing

The router works with 15 task types. A request's task type is the category that the embedding and the classifier together assign to it: a bge-small embedding represents the request as a vector, and a linear layer on top of it estimates the probability $p(t)$ of each task type t. For every combination of task type and model, observed pass rate and mean billed cost per request were calculated. Weighting these estimates by the classifier's task probabilities gave an expected pass rate and cost for each candidate model m. Models were ranked using the following score:

$$\text{Routing score}(m) = \sum_t p(t)\,\text{accuracy}(t, m) \;-\; \lambda \sum_t p(t)\,\text{cost}(t, m) \tag{3.1}$$

Here $\text{accuracy}(t, m)$ is model m's observed pass rate on task type t, and $\text{cost}(t, m)$ is its mean billed cost per request in USD. Because both terms are weighted by the task probabilities, an uncertain classification gives a smooth decision instead of a hard mistake. The parameter λ controls how much weight the router gives to cost. Higher values favour cheaper models. The default configuration used $\lambda = 20$. Earlier routing rules and classifier comparisons are described in Appendix B. Results are reported for $\lambda = 20$ and $\lambda = 10$, and for the full range of λ in Appendix B.4.

Evaluation

Each routing strategy was evaluated by replaying its decisions against the stored outputs. When a strategy selected a model, that model's recorded output and cost determined the outcome for the request. This allowed successive versions to be compared without generating new outputs. The final router was also evaluated with its answer check enabled and disabled, to measure the change in pass rate and the cost of additional calls.

Five-fold cross-validation was used at the prompt level to keep each test prompt's outcomes out of its routing-table estimates. All eight outputs for a prompt stayed in the same fold. In each round, the routing table was built from four folds and evaluated on the fifth. Each fold served as the test fold once, and the procedure was repeated across ten random partitions. Figure 3.1 illustrates the procedure.

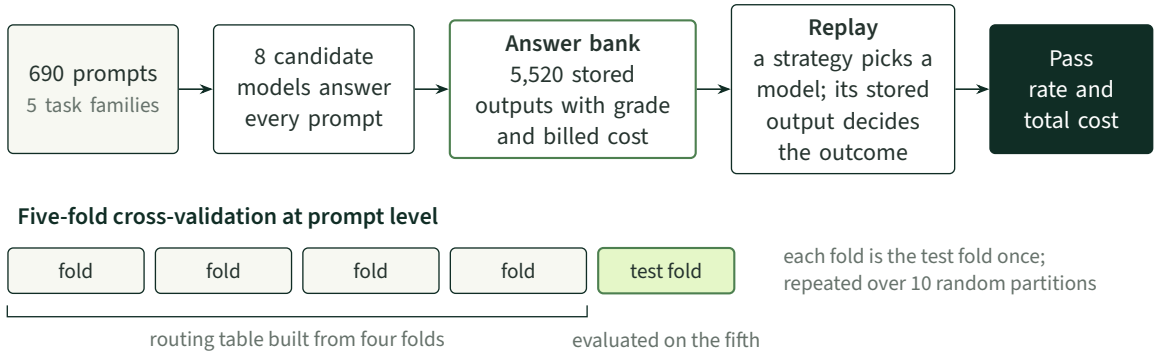


Figure 3.1. Replay evaluation. Every model answers every prompt once; routing strategies are then compared on the stored outputs, with the routing table always built from prompts other than those it is tested on.

The main measures were the overall pass rate across all five task families and total provider cost, including additional calls. The router was compared with two reference points. The strongest fixed model was the single model with the highest overall pass rate when used for every prompt. For each prompt, the benchmark oracle selected the cheapest model with a passing output. Where no model passed, it selected the cheapest model. The classifier processing time was measured separately on a CPU.

3.5 Financial analysis

The financial analysis examined two questions: how much of the benchmark saving a customer would retain after service fees, and how much usage Lönn would need to cover its operating costs. Gross savings were first calculated as the difference between the fixed reference model's cost and the router's provider cost for the same prompts. The proposed service fee was then applied to those savings to estimate customer charges and Lönn's revenue.

To estimate break-even on non-personnel operating costs, variable costs were deducted from fee revenue to give the contribution per dollar of underlying provider spend. Monthly fixed costs were divided by this contribution to obtain the required monthly provider spend. The usage estimate from the customer consultation was then used to express break-even as an indicative number of customers. Pricing, customer usage and operating costs were treated as scenario assumptions.

4 Assumptions and requirements

The pre-study examines three conditions for Lönn AI's feasibility. The service must create enough value for customers to justify its fee, select models reliably enough to make routing worthwhile, and handle requests under conditions customers can accept. The assumptions below state what must hold for the concept to be viable. The requirements describe what the proposed product must deliver.

4.1 Business

Assumptions

A1. Organisations using LLM APIs incur sufficiently high inference costs for intelligent routing to provide meaningful financial value, through measurable savings and greater cost transparency.

A2. Part of the value created by intelligent routing can be captured through a pricing model that customers accept and that creates a sustainable profitable business.

Requirements

R1. Lönn AI must deliver measurable financial value to customers after accounting for its service fees.

R2. Customers must have visibility into and control over their AI spending, including usage breakdowns, cost forecasts, and routing preferences.

R3. Lönn AI must have a pricing and payment model that captures a sufficiently profitable share of the savings it creates while limiting its credit risk towards LLM service providers.

The commercial viability of Lönn AI depends on two things, whether the savings and cost transparency it provides are valuable enough to matter to customers, and whether part of that value can be captured in a sustainable business model. The first assumption is that LLM API costs have become a problem for organisations, both in size and in transparency. An article by Clari (2026) shows that building an affordable prototype can be easy, but keeping a useful product affordable once it receives real traffic is more difficult. With usage-based pricing, this concern seems like common sense in the market. Costs are also hard to follow with many different LLM API providers (e.g. Anthropic, OpenAI) (DeltaLLM, 2026). Companies can easily lose track of spending spread across separate cost-tracking portals. Lönn AI should therefore give customers not only measurable savings, but also insight into and control over their LLM API costs.

The second assumption is that this value can be captured. Solving the cost problem only generates revenue if Lönn AI can take a share of the savings while customers remain better off after its fees. OpenRouter, for example, charges a platform fee of 5.5% on credit purchases (8% on its business tier) (OpenRouter, 2026). A similar percentage fee could be paid either through prepaid credits or through after-the-fact invoicing. This choice also determines Lönn AI's credit risk. As a middleman between LLM providers and customers, it must avoid being stuck with a large provider bill that a customer cannot or will not pay. Finally, this revenue stream from the share of savings must be large enough to provide a profitable and sustainable business.

4.2 Technical

Assumptions

A3. LLMs in the model pool have complementary strengths that are sufficiently predictable for per-request model selection to achieve a better cost–quality trade-off than using a single fixed model across all requests. These capabilities can be estimated and updated as models enter or leave the pool.

A4. The computational cost and latency introduced by routing are sufficiently low to preserve the benefits of dynamic model selection.

Requirements

R4. Lönn AI must reduce overall inference costs while maintaining comparable output quality and acceptable latency with minimal integration effort.

R5. Lönn AI must ensure reliable and maintainable routing through automated fallback mechanisms when providers become unavailable and modular updates when new models or price changes are introduced.

The technical feasibility of Lönn AI depends on its ability to identify which model is most suitable for a given request. By evaluating characteristics of the request, the router aims to direct simpler requests to lower-cost models while reserving more capable models for demanding tasks.

Beyond model selection, the routing process must remain computationally efficient. Additional processing introduces both latency and operational costs, which could offset the savings achieved through selecting cheaper models. Therefore, the benefits of intelligent routing must be evaluated against the overhead introduced by the system itself. Furthermore, reliable request execution is essential for practical deployment. As model and provider availability can vary, the system must support fallback mechanisms that redirect requests when the preferred execution option is unavailable.

4.3 Trust and safety

Assumptions

A5. Lönn AI's customers will accept standard data-handling settings and documented safeguards as sufficient assurance to use the service.

A6. Provider configurations and contracts can be verified sufficiently to enforce these settings across the eligible execution pool.

Requirements

R6. Lönn AI must provide standard settings for processing location, content retention and training use, including zero data retention, and enforce them across its own systems and downstream providers.

R7. Lönn AI must maintain appropriate security controls, standard confidentiality and data-processing agreements, current provider information, and documented retention, deletion and incident procedures.

Lönn's trust requirements centre on protecting customers' confidential business information, which may include commercially valuable trade secrets (European Commission, n.d.). Security controls and contractual commitments must govern access, disclosure, retention and training use across Lönn and its model providers.

Lönn should offer all customers predefined settings for processing location, retention and training use. Providers such as OpenAI (n.d.-b) and Anthropic (n.d.-c) publish retention policies and zero data retention (ZDR) conditions for their application programming interfaces (APIs). Lönn is designed to match these conditions and verified endpoint configurations against the customer's settings to determine eligibility. Its ZDR setting requires prompt and response content to be excluded from persistent storage throughout the processing route, making eligibility dependent on the endpoint and features used (OpenAI, n.d.-b; Anthropic, n.d.-c).

Verification continues after integration. Lönn would maintain a rule-based execution pool, applying eligibility rules to every request and revalidating provider and model configurations periodically and following relevant changes.

The International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC) publish ISO/IEC 27001:2022, a standard for information security management. Certification is optional (ISO, n.d.). Lönn's initial approach therefore focuses on documented controls, processing agreements and implementation evidence. Customer acceptance of this assurance remains an assumption requiring validation.

5 Market analysis

5.1 LLM routing market

LLM routing as a commercial category is young. A reasonable starting point is 2023, when OpenRouter and Martian described their products, LiteLLM launched and Cloudflare introduced AI Gateway (Martian, n.d.; OpenRouter, n.d.; Y Combinator, n.d.; Cloudflare, n.d.-a). At that stage the focus was on making it easier to reach many providers through one interface. From 2024 and 2025 the focus moved toward choosing a model based on the request itself, for example through AWS Intelligent Prompt Routing, Microsoft Model Router and Vercel AI Gateway, and routing started to become part of mainstream cloud infrastructure (Amazon Web Services, 2024; Microsoft, n.d.; Vercel, n.d.). In 2026 the offers are more configurable and enterprise-oriented, with explicit control over cost and quality (Microsoft, n.d.), budgets and quotas (Cloudflare, n.d.-b), and regional controls for where inference runs (Oracle, n.d.). Taken together, these developments suggest that competition is moving from giving access to many models toward deciding which model and provider should handle each request, within limits for cost, performance, reliability and governance.

5.2 Competitors

The competitors can be divided into four segments, based on differences in product focus and in how the product is deployed. The division gives a clearer view of who the main competitors are, what they offer and where the competition is concentrated.

Table 5.1. Competitor segments by product focus and deployment model.

Segment	Definition	Key competitors
Multi-model routing platforms	One API to many models and providers, with routing based on quality, cost, latency and availability	OpenRouter, Martian, MegaRouter, Requesty, Not Diamond, Vercel AI Gateway, Eden AI
Enterprise AI gateways	Central control of AI traffic: governance, security, budgets, observability and routing	Portkey, TrueFoundry, Cloudflare AI Gateway
Self-hostable AI gateways	Gateway and routing software that runs in infrastructure controlled by the customer	LiteLLM, Kong AI Gateway, Helicone, Bifrost
Cloud-native AI routing	Routing built into an existing hyperscaler environment	Microsoft Foundry, AWS Bedrock, Google Vertex AI, Oracle OCI

Lönn AI mainly targets the multi-model routing platform segment, where the competition is about selecting and reaching the most suitable model and provider for each request. The adjacent segments still matter. Many of them, especially the cloud-native platforms and some enterprise gateways, are limited to the models, providers and infrastructure supported inside their own ecosystems. As these platforms add more models and more routing features, they could overlap more with the multi-model routing market and change the competitive picture.

5.3 Positioning

Lönn AI should be positioned between the broad model access that platforms such as OpenRouter offer and the more intelligence-driven routing approach of companies such as Martian. The aim is not to compete on the number of models and providers. The aim is to choose a sufficiently broad and relevant portfolio of models and to optimize how each request is carried out. The core of the differentiation is cost-efficient model selection that keeps output quality. This is achieved through model and provider selection using current research in LLM routing and inference optimization. The goal is not hundreds of possible routes. It is better decisions across a diverse and complementary chosen set of models and providers. A second part of the differentiation is the web application. Its dashboard, with spending forecasts and usage breakdowns by team, task and model, and its settings for data-handling policies give customers visibility and control alongside the routing (Section 8.2).

This position fits a market where model capabilities, provider prices and infrastructure costs change quickly. Lönn AI should therefore act as a dynamic optimization layer between customers and providers and keep adjusting its routing as the market changes. Besides lowering cost, this can give customers more stability, since quality stays consistent and AI spending becomes easier to predict.

6 Stakeholders

Customers. The customer is a company that uses LLM provider APIs and pays for the tokens, either inside a product it sells or in its own internal tools and workflows. Within the company there are two roles. The people who decide to adopt Lönn AI and who carry the cost are usually found in the company leadership or the budget owner, such as a CTO, a head of engineering or someone responsible for the IT budget, depending on the size and profile of the company. The developers integrate the API and see the effects in practice, but they do not normally make the purchasing decision. The two roles want different things. Leadership wants lower and more predictable cost with maintained answer quality. Developers want an integration that is small and does not change how the application behaves.

LLM providers. The providers are the suppliers the product depends on for every request. Without access to their models there is no product. This creates a dependence that must be managed. A pricing change, a limit on access or a new competing feature from a large provider can change the conditions for the product quickly. One reason to use several providers is to reduce this dependence.

Regulators. GDPR (European Union, 2016) and the EU AI Act (European Union, 2024) shape what data can pass through the product and how it can be stored. Since the product handles prompts that can contain company data and personal data, the question of where data is processed and whether it is stored is central. This affects both the technical design and how much trust customers are willing to give the product. These rules are enforced by national data protection authorities, in Sweden the Swedish Authority for Privacy Protection (IMY), and, for the AI Act, by national market surveillance authorities and the EU AI Office. Customers in regulated sectors may bring further sector-specific requirements. The implications for Lönn AI are discussed in Section 9.3.

Competitors. The competitors are the companies that offer routing and gateways for LLM APIs. As an API middleware, Lönn AI competes with them directly for the same customers. They are described in more detail in Section 5.2. Large cloud providers that build routing into their own platforms are also part of this group.

Investors. Investors are not engaged at the pre-study stage, but they matter for whether the venture can grow beyond a prototype.

7 Participatory design interviews

7.1 Product development

To test the concept against a real use case, an interview was held with a founding engineer at Zaplar, a startup accelerated by Y Combinator. Zaplar builds an agentic hotel system in which several features call LLM APIs, and the engineer works on its full-stack development. The company is therefore the kind of customer described in the scope: it relies on LLM APIs inside its product and pays for every request.

The engineer considered the product highly relevant, with potential in many areas. The strongest fit was in parts of a system where one agent handles a varied mix of easy and difficult tasks, or where agents fetch answers from different models for individual tasks. Examples were tool calling,

natural-language writing, code writing and computational tasks. Because these requests differ in difficulty, sending each one to a suitable model instead of one fixed model is where routing creates value.

The engineer also pointed to a limit. Some functions require a fixed level of output quality, and there predictable results matter more than a lower cost. How useful routing is therefore depends on the specific implementation, and the customer needs a way to decide how quality and price should be balanced.

This feedback is reflected in the final solution. The router's λ parameter, described in Section 3.4, sets how much one dollar per request is worth in accuracy, so each customer can move the balance towards lower cost or higher quality. A function that needs consistent quality can use a low λ, while simpler high-volume tasks can use a higher one. The interview also supports the business requirement that customers must have control over their routing preferences.

7.2 Trust

To build customer trust and ensure data security, Lönn AI's trust and safety approach was developed iteratively under the mentorship of an operations specialist with experience from an AI startup that processes sensitive health-related information in its products. The specialist highlighted several key measures: working towards the appropriate ISO certification, ensuring compliance with the GDPR, enforcing a 30-day retention limit, hosting all data within the EU, and applying encryption at rest. Encryption at rest was described by the specialist as data being encrypted while it is stored, as opposed to only when it is in transit. In addition, the specialist explained the requirements of the EU AI Act, and an approach was developed together with the specialist for conducting risk assessments based on the probability and impact of potential risks.

This collaboration with a specialist working in a sector with some of the strictest data privacy requirements has shaped Lönn AI's approach to data privacy. Finally, drawing on their market knowledge and professional experience, the specialist explained that actively working with the factors described above is adequate to build the customer trust needed for customers to trust a start-up with their data.

7.3 Customer insight

To get insight into the size and structure of Lönn AI's potential customers, a person working at a company who would be a potential customer was consulted. This person will remain anonymous within this report. The main question was how much usage the average customer would be expected to have, in terms of LLM tokens. The interesting token count here would be the number of tokens the potential customer uses within their products, targeting LLM provider APIs, not their total usage in all tools. The person described that this should be expected to vary a lot depending on the type of product and the number of users the potential customer has. However, for potential customers developing a B2B solution with approximately 10 users (other companies), the person estimated that 100 million to 1,000 million tokens per month would be required to operate it.

8 Results

8.1 Evidence from the routing literature

Fowler et al. (2026) found that models produced correct outputs on different subsets of benchmark tasks. Across 21 LLMs and 16 benchmarks, the oracle achieved an average score of 88.8%, compared with 82.4% for the strongest fixed model on each benchmark. The improvement shows that some errors made by the strongest model were recoverable through model selection, supporting the assumption in A3 that capabilities vary across tasks. The authors also examined cost at a fixed level of quality. When the oracle matched the strongest fixed model's score, it reduced token costs by 85.2% on average.

An earlier, separate study by Ong et al. (2025) addressed a related question: whether differences between models are predictable enough to guide model selection in practice. Their system, RouteLLM, used preference data to learn which requests benefited from a stronger model. On MT-Bench, routing between GPT-4 and Mixtral 8x7B retained 95% of GPT-4's benchmark score at an estimated 73% lower inference cost. This supports the predictability component of A3. The authors also estimated that even their most computationally expensive router added at most 0.4% relative to the cost of GPT-4 generation, providing evidence for the low computational overhead assumed in A4.

Li et al. (2026) provide direct evidence for the cost and quality components of R4. On LLMRouterBench, Avengers-Pro (Zhang et al., 2025) achieved at least the average accuracy of GPT-5, the strongest fixed model in the comparison, at 31.7% lower inference cost. In a separate configuration that prioritised quality, Avengers-Pro achieved 4.0% higher average accuracy relative to GPT-5. These results demonstrate two attainable outcomes of routing: preserving benchmark accuracy at lower cost and improving accuracy by using the complementary capabilities of several models.

Yan et al. (2026) examined how a router can adapt as providers release new models. Their system, ZeroRouter, represented task difficulty and model capabilities in a shared space, enabling it to route requests to models outside its original training pool. This provides evidence for a feasible approach to model onboarding in Lönn's technical roadmap: the available model pool can be updated without retraining the entire router after each release.

8.2 Proposed product design

Lönn is a middleware that sits between customer applications and model providers as an optimisation layer. Its system design connects customer controls, request routing, and evaluation and improvement, designed to address the business, technical and trust requirements in one product.

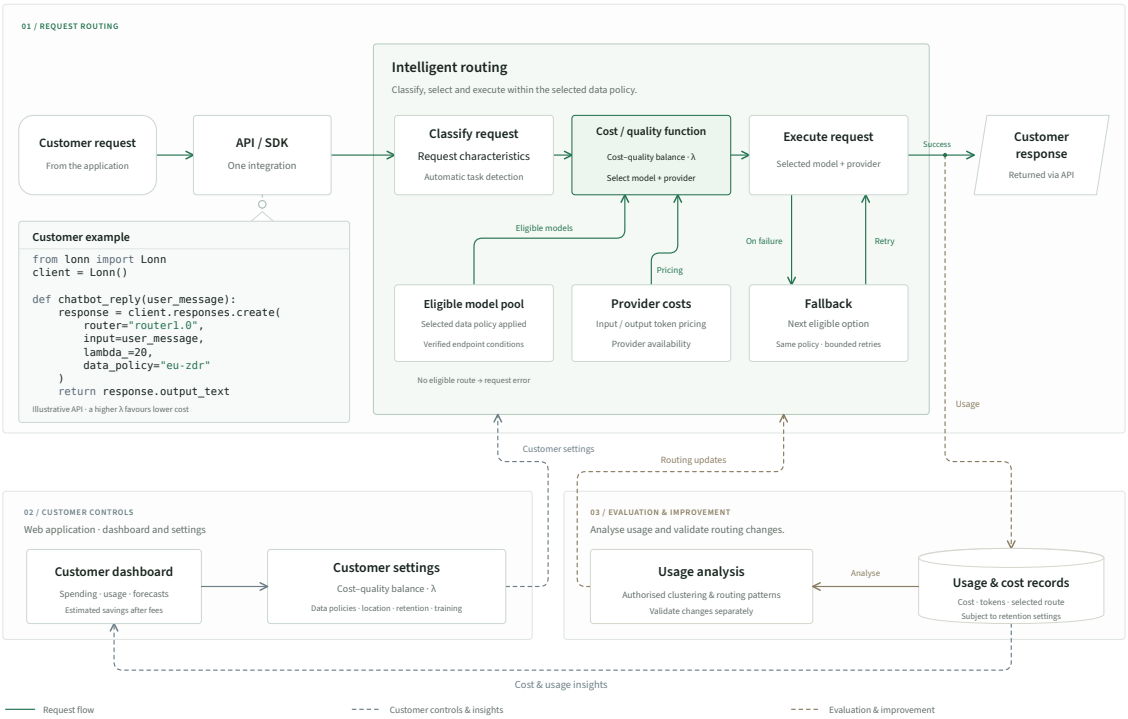


Figure 8.1. Proposed system architecture.

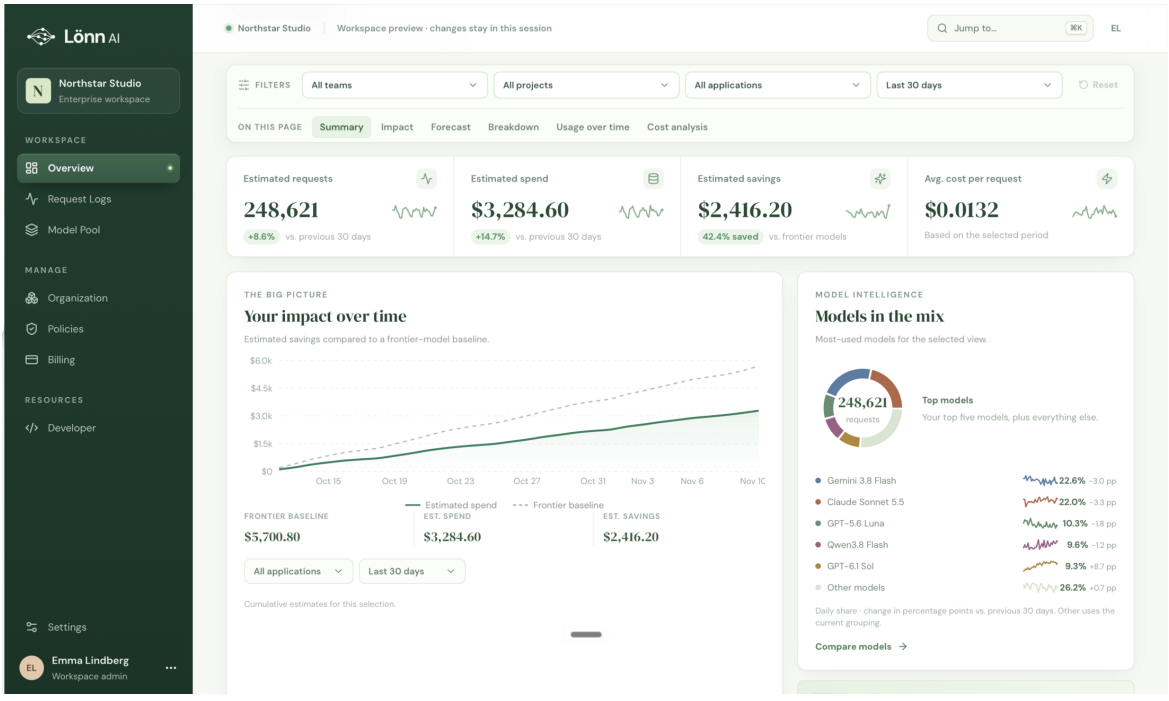


Figure 8.2. Concept mock-up of the customer dashboard.

Customer controls

Customers integrate through an API and Python software development kit (SDK), specifying a router version. The web application includes a dashboard and settings. The dashboard presents spending forecasts, usage breakdowns by team, task and model, and estimated savings after service fees. Settings define data-handling policies for processing location, retention and training

use, including zero data retention. Each call can select a saved policy through data policy and set the cost–quality balance through λ.

Request routing

The router automatically identifies request characteristics and selects the most cost-efficient eligible model and provider conditioned on λ. The selected policy and regularly verified endpoint conditions define the execution pool. Fallback follows the same restrictions within retry limits. Requests fail when no eligible route remains. Security controls, contractual safeguards and documented procedures protect confidential information.

Execution records support billing and savings estimates comparing customer charges, including service fees, with predicted frontier-model costs. Prepaid balances or credit limits constrain exposure to unpaid provider charges.

Evaluation and improvement

Authorised retained prompts support unsupervised clustering and analysis of routing behaviour. Retention and reuse follow customer permissions. Validation must establish comparable answer quality and acceptable latency and assess net savings after service fees and routing overhead.

8.3 Lönn Router prototype and performance

The Lönn Router consists of a classifier, a router and an executor. The classifier identifies the likely task type, the router ranks eight candidate models by expected quality and cost according to the customer's preference, and the executor calls the selected model and records the cost of each attempt. If a provider call fails, the executor tries the next ranked model. If a call succeeds but its output is empty, truncated or fails a code example supplied in the prompt, an answer check can trigger one additional call. Figure 8.3 shows the flow.

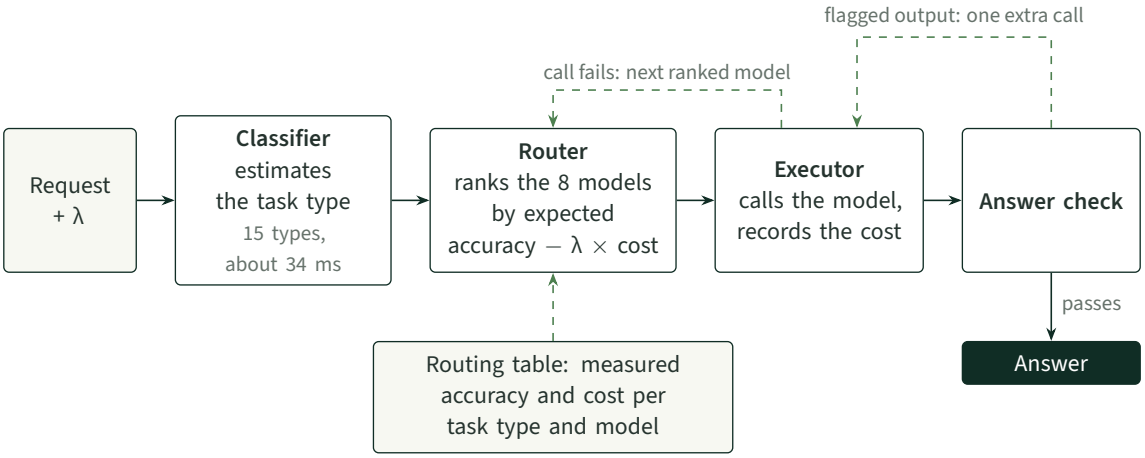


Figure 8.3. How the Lönn Router prototype handles a request. Dashed arrows are the two recovery paths: fallback when a provider call fails, and escalation when the answer check flags the output.

The Lönn Router answered 85.8% of the benchmark prompts correctly at a cost of 1.58 USD, against 88.1% at 7.58 USD for always using Claude Sonnet 5.5. It thus keeps about 97% of Sonnet's accuracy at about a fifth of its cost. With λ = 10 it answered 89.6% correctly at 3.91 USD, which

means higher accuracy than always using Sonnet at about half the cost (Table 8.1, Figure 8.4 and Appendix B.4).

Table 8.1. Main results on the 690 benchmark prompts (cross-validated). Costs are total provider costs for all prompts.

Configuration	Correct	Cost (USD)	Compared with always using Sonnet
Lönn Router without answer check, λ = 20	83.7%	0.91	−4.4 points, −88% cost
Lönn Router, λ = 20	**85.8%**	**1.58**	−2.3 points, −79% cost
Lönn Router, λ = 10	**89.6%**	**3.91**	+1.5 points, −48% cost
Always Claude Sonnet 5.5	88.1%	7.58	–
Oracle	94.1%	1.08	–

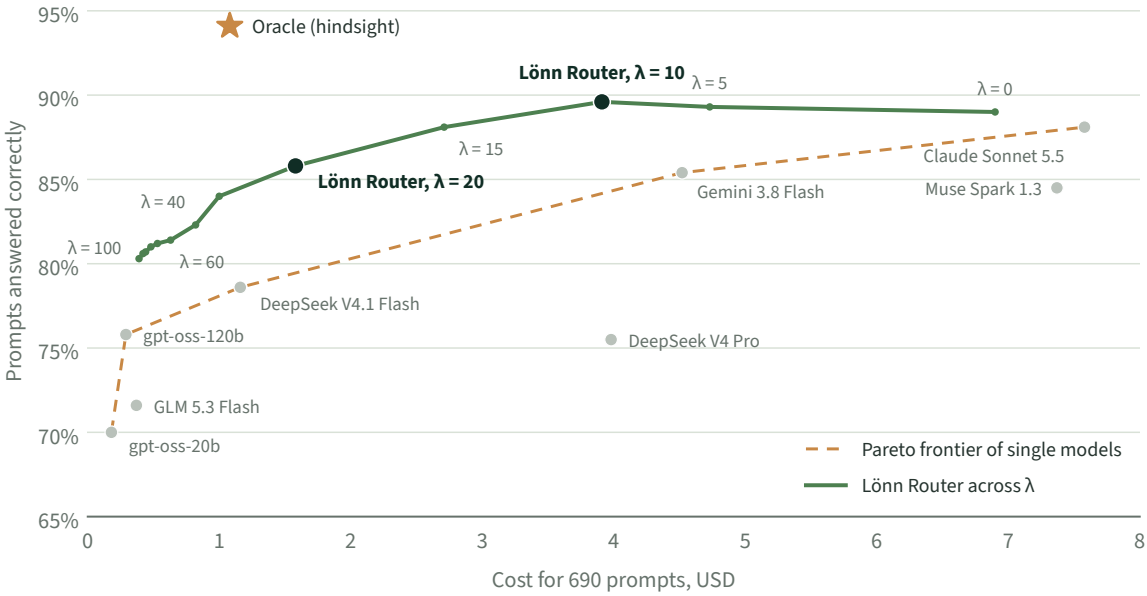


Figure 8.4. Accuracy against cost for the eight single models and the Lönn Router across λ.

These results also support the technical assumption A3: a practical router can reduce cost substantially while keeping quality close to, or above, that of a strong model. These results apply to the requests included in the benchmark. The cost savings and output quality achieved on customer traffic may differ depending on the mix of tasks, costs and quality requirements in each application.

8.4 Profitability analysis

To assess the profitability of the Lönn AI concept, a financial model has been built as a part of this pre-study. The model was developed to determine which financial conditions had to be fulfilled for the potential business to be profitable. It is based around the Lönn AI benchmark, assuming customers previously use Claude Sonnet 5.5, and are sized according to how they were described in the participatory design interviews (Section 7.3). This model breaks down the Lönn AI startup's potential financials into an income model, a variable costs model, a fixed costs model, and to

summarize it, a profit model calculated from all the previously mentioned models. For the full model, see Appendix D.

First, the income model calculates the potential income per routed USD of AI spend, based on a pricing model where Lönn AI charges a service fee of 20% of what the customer saved by using the Lönn router for their request. This is the proposed pricing model for the product, combined with using pre-paid credits which require customers to pay in advance of usage. The 20% service fee results in Lönn AI receiving 0.759 USD of income per routed USD of AI spend used by customers. Figure 8.5 shows how the customer's bill is composed in the benchmark case.

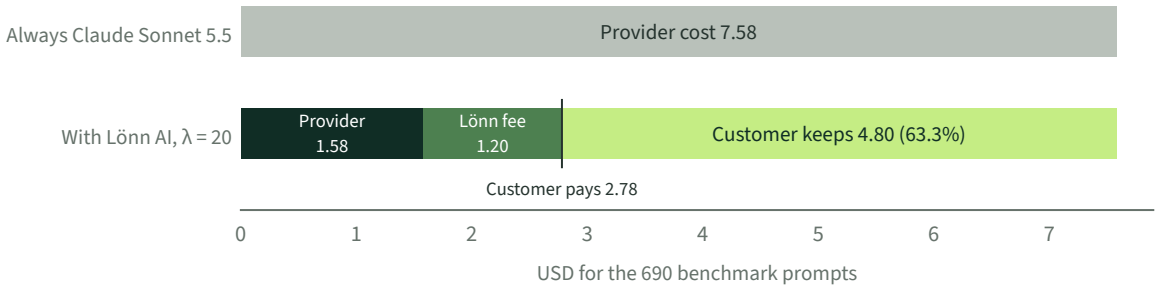


Figure 8.5. The benchmark case from the income model: what the customer pays with and without Lönn AI. Of the 6.00 USD gross saving, Lönn AI keeps 1.20 USD (20%) and the customer keeps 4.80 USD.

Secondly, the variable costs model calculates the variable costs of the startup on a per routed USD of AI spend basis. These costs mostly include payment-provider and conversion related fees. The calculation within the model estimates the variable costs to 0.047 USD per routed USD of AI spend.

Thirdly, the fixed costs model covers essential monthly recurring and one-time fixed costs such as hosting, development tools and administrative expenses. The development costs up until this point are however not included, as they are considered sunk. This model results in 421 USD per month in fixed monthly costs, as well as 241 USD in one-time startup costs.

Finally, in the profit model, the contribution margin is calculated to 0.713 USD per routed USD of AI spend and the contribution margin % is 93.8%, based on the previously described calculations and sub-models. This results in a break-even of 592 routed USD of AI spend per month, for the Lönn AI concept to be profitable. This is roughly the equivalent of 2 customers depending on their business and size. Figure 8.6 shows profit as a function of routed spend.

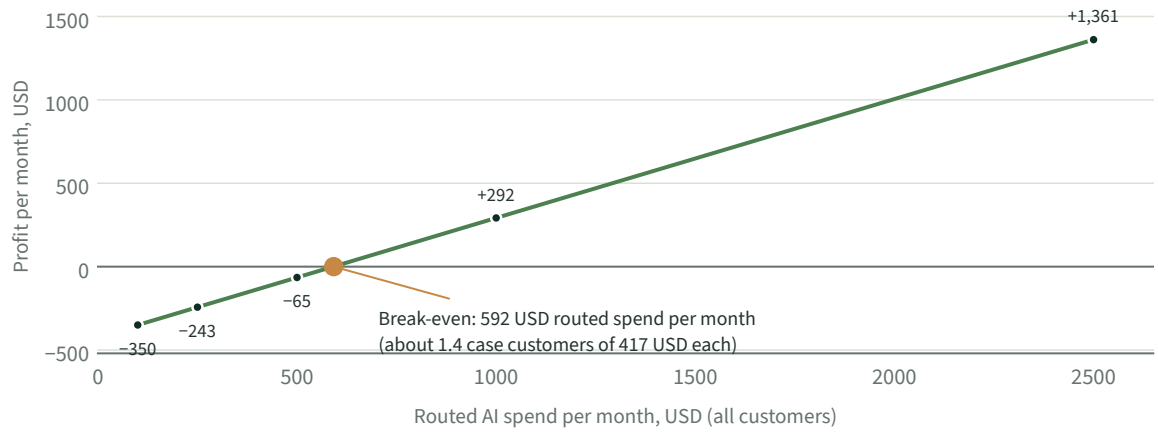


Figure 8.6. Profit per month by monthly routed AI spend, from the financial model (Table D.3). Fixed costs are 421 USD per month and exclude salaries.

9 Discussion

9.1 Business requirements (R1–R3): partially met

The business requirements from the assumptions and requirements part of the report specify that Lönn AI must deliver financial value to customers after accounting for service fees, provide visibility and control into their AI spending, and have a pricing and payment model that captures a profitable share of this value, while limiting credit risk. Based on the benchmark, the router delivers clear financial value, and at $\lambda = 20$ it cuts AI spend by 63.3% after accounting for the service fee, according to the financial model, for the assumed typical customer. Additionally, this model is based on a pricing and payment model where Lönn AI takes a 20% cut of the customer's AI savings, as a service fee. This pricing model, combined with pre-paid usage credits provides profitability after only 2 assumed typical customers, while removing most credit-risk.

Furthermore, the λ variable gives the customer direct control over the balance between cost and quality, which the Zaplar interview confirmed the need of. In terms of visibility, a dashboard has been designed which provides usage breakdowns on relevant groupings. This dashboard would also provide a single point of billing and usage statistics for the customer's total AI spend across all providers.

However, the business requirements should only be considered partly met since the evidence is derived from the benchmark and individual interviews. The savings are calculated based on an assumed typical customer, no customer has yet been asked whether they would accept the fee, and the size of the savings depends on the mix of tasks that the customer is running through the system.

9.2 Technical requirements (R4–R5): R4 partly met on the benchmark, R5 not yet met

The technical requirements specify that Lönn AI must reduce overall inference costs while maintaining comparable output quality and acceptable latency. They also specify that routing must be reliable and maintainable, through automated fallback when providers become unavailable and modular updates when models or prices change. Based on the benchmark, the router keeps about 97% of Claude Sonnet 5.5's accuracy at about a fifth of its cost at $\lambda = 20$ (before the service fee), and at $\lambda = 10$ it answers more prompts correctly than Sonnet at about half the cost. The classifier takes 34 ms per prompt on an ordinary CPU, which is small compared with typical model response times. On the benchmark's requests, this supports the assumptions that models have complementary strengths that can be predicted (A3) and that routing adds little cost and latency (A4), and it agrees with earlier research such as RouteLLM and Avengers-Pro. Assumption A3 is stated for all requests, however, and has been tested only on the task types in the benchmark. End-to-end latency, including the extra call for escalated requests (8% at $\lambda = 20$), was not measured, and neither was integration effort.

In terms of reliability, the prototype falls back to the next model in the ranking when a provider call fails, and the routing table can be rebuilt when prices change. Adding a new model, however, currently requires running it through the full benchmark, and fallback has not yet been tested under realistic load. R5 is therefore not yet met.

The cost and quality part of R4 should therefore be considered met on the benchmark, with three main reservations. First, the dataset is limited to 690 prompts with one stored answer per model, and a larger dataset would allow more certain statements. Second, the benchmark consists of exam-style questions, which differ from the prompts customers send in practice. The WildChat test in Appendix B.2 confirms this: the classifier identified the correct task type for only 51% of everyday prompts. Third, the task types, the answer check and the λ values were chosen on the same benchmark that produced the results. Usage data from real implementations, and training the classifier and routing table on that data, would give results that better reflect how the router performs in production. Even so, the cost difference compared with always using Sonnet is large across the whole range of λ, which suggests that substantial savings are likely on real traffic, although their size remains to be shown in the pilots. Latency for escalated requests, the effect of model switching on caching, and an improved pipeline for onboarding new models are left for further work (see Section 10.1).

Supplier risk

A further technical risk lies outside the router itself, in the dependence on the model providers. In Lönn AI's set-up the provider becomes interchangeable, and the customer may not know which model answered. A provider that sees this as a loss of brand recognition or customer relationships can raise prices or restrict access. Anthropic's cutting of Windsurf's access to Claude in 2025 (Zeff, 2025), and Anthropic acting as both Cursor's supplier and competitor (Garfinkle, 2026), show that this happens in practice. A model pool spread across several providers reduces this risk, but losing access to the strongest models would lower the value of the whole product.

9.3 Trust requirements (R6–R7): partially met

The trust and safety requirements specify that Lönn AI must provide settings for processing location, content retention and training use, and ensure that these are enforced at downstream providers. Additionally, the requirements state that Lönn AI must maintain appropriate security controls, including confidentiality, data-processing agreements, updated provider information, and documented procedures for data retention, deletion and incidents. These policies and documentation have been discussed and planned in cooperation with an operations specialist at an AI start-up that processes sensitive health information, however, they have not been developed in practice. Turning these plans into concrete policies and practices is therefore a roadmap item for Lönn AI. The measures the specialist highlighted are listed in Section 7.2. R6 is therefore supported in theory but not validated, and R7 is judged feasible but has not been tested in practice.

As a middleware that forwards prompts, Lönn AI would act as a data processor for its customers under the GDPR (European Union, 2016), with the model providers as sub-processors. Routing to providers outside the EU would require a legal basis for international transfers, or exclusion of those providers from EU-only policies.

Additionally, data security comes as a trade-off against customer AI spend savings. With stricter data processing requirements, fewer AI providers will naturally be available in the model pool, and several of the cheaper models might then be excluded. Because of this, customers with very high data processing restrictions might receive significantly lower savings than what is estimated based on the benchmark.

9.4 Limitations

The results should be read with several limitations in mind. The benchmark consists of 690 exam-style prompts with one stored answer per model, so small differences between configurations are uncertain. The writing tasks were graded by an LLM judge, which may favour some models. The oracle is an optimistic upper bound because it benefits from answers that were correct by chance. The participatory design evidence comes from three individual interviews. Finally, the data-handling safeguards, cost attribution and dashboard are designs that have not yet been implemented or tested.

9.5 Final remarks

Returning to the question that opened the solution, whether a company could stop choosing models altogether, this pre-study gives a clear answer on the benchmark: routing lowers the cost of LLM use while keeping answer quality close to that of a strong model, the prototype shows that the approach can be implemented, and the business can cover its non-personnel operating costs with a few customers. What remains is to show the same on real customer traffic, and the roadmap below sets out how Lönn AI will get there.

10 Roadmap

10.1 Technical roadmap

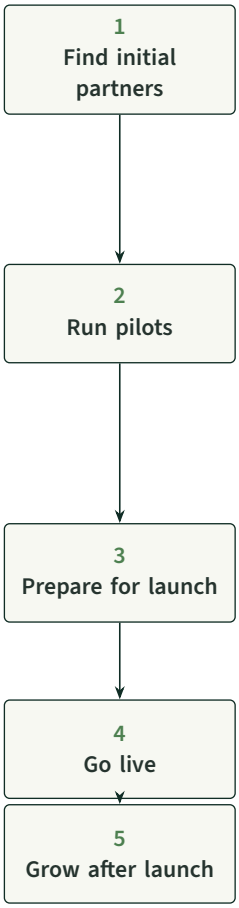
Table 10.1. Technical development beyond the scope of this pre-study.

Workstream	Next development
Model routing	Further development should make it straightforward to onboard new models without retraining the complete routing system. Model capability vectors should be learned from observed performance data rather than manually engineered qualitative descriptions. One-shot routing, in which a new model is profiled on a small set of anchor tasks (Yan et al., 2026), is one possible approach to investigate. A small set of shared anchor tasks could provide evidence for estimating new model vectors under controlled conditions. Further work must establish how much representative data is needed for reliable onboarding and model selection.
Evaluation	Broader, representative evaluation data and repeatable tests are needed to establish whether routing preserves answer quality while reducing total costs and maintaining acceptable latency. Authorised customer usage can support clustering and workload analysis, while answer quality requires separate validation.
Trust and data handling	Customer policies for processing location, retention and training use must be enforced across Lönn and downstream providers. Provider conditions and endpoint configurations require ongoing verification. Security controls, contractual safeguards and procedures for retention, deletion and incident handling also require implementation.

Workstream	Next development
Reliable execution	Provider integrations, failure handling, fallback and monitoring require further development and testing under realistic workloads. Retries must remain within the selected data policy and defined limits.
Customer controls and integration	The API, Python SDK and web application must connect individual requests with their cost–quality preferences and saved data policies. The dashboard requires spending forecasts and usage breakdowns, while settings must be applied consistently throughout routing and execution.
Metering and billing	Accurate usage tracking, current provider prices and payment controls are needed to support billing and limit financial exposure. Savings reports must distinguish measured customer charges, including service fees, from predicted frontier-model costs. Prepaid balances or credit limits require technical enforcement.
Caching and session memory	Further work should assess how model switching affects key–value (KV) caching and session memory. Relevant conversation context must remain available across requests and model changes while respecting customer data policies. Lost caching benefits and the cost of passing additional context must be considered when estimating routing savings.
Prompt and reasoning optimisation	Prompt compression and prompt engineering should be investigated as additional ways to reduce token consumption. Reasoning limits adapted to request complexity could also reduce output tokens. These approaches require validation to determine whether savings can be achieved while preserving answer quality.

10.2 Business roadmap

The business roadmap is a plan of action rather than a schedule. It sets out the order in which Lönn AI moves from a prototype to a product that anyone can use.



Find initial partners. Recruit a small number of partner companies that rely on LLM APIs in their products, for example AI-heavy startups in Stockholm such as Legora or Lovable, or growing AI startups such as Zaplar. During the pilots, partners pay only the underlying model costs and no service fee. In return, they share usage data on agreed terms, give feedback on the product and allow Lönn AI to name them as partners.

Run pilots. The pilots aim to collect usage data from real traffic, gain insight into what customers need from the product, verify that the product works under real conditions, and build credibility through known partners. Each pilot starts in a shadow mode, where the router only records which model it would have chosen, before it gradually takes over the partner's real traffic.

Prepare for launch. Build the data and insights from the pilots into the product, so that it is ready for public use. At the end of the pilots, partners move to the standard pricing model and become the first paying customers.

Go live. Open Lönn AI to everyone. Any company can create an account, receive a Lönn AI API key and route its requests through it.

Grow after launch. Use the partners as references to win similar customers, and use the growing amount of usage data to keep improving the router.

Figure 10.1. The five steps of the business roadmap.

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A Disclosure of AI use

Within this pre-study, AI has been used in accordance with the guidelines given by the course examiners. AI has always been used with a human driving the overarching work process and has simply been used as a tool to effectively carry out tasks. More specifically, AI has been used for reformulation of parts of text with the purpose of improving grammar and flow, Excel workbook building assistance (with humans making all decisions and providing constant direction), and coding.

B Earlier iterations

This appendix describes the earlier iterations of the routing core in more technical detail: what was tried, what the evaluation showed and what was learned. The process behind them is described in Section 3. Figure B.1 gives an overview of the path.

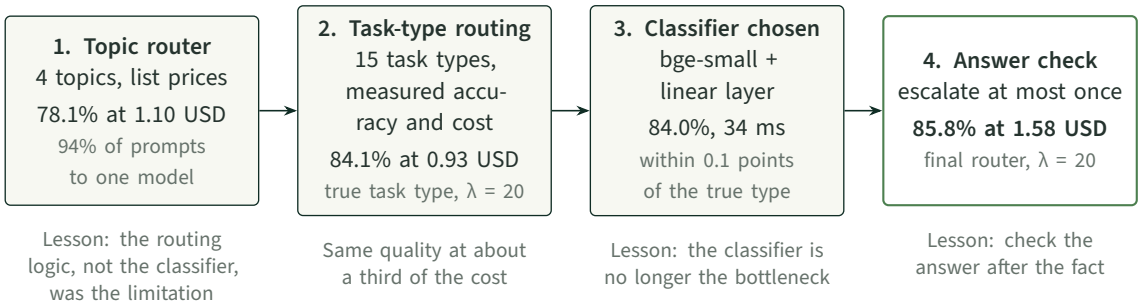


Figure B.1. From the first topic router to the final Lönn Router: accuracy and cost on the 690 benchmark prompts at each step, with the lesson drawn (Tables B.2–B.7 and Table 8.1).

B.1 Topic router

The first iterations rested on the hypothesis that a router can choose well if it knows what a prompt is about and how hard it is. The analysis step used Laya, an open general-purpose text classifier based on ModernBERT, which answers questions about a text that are posed in natural language. It assigned each prompt to one of four topics (coding, maths and reasoning, creative writing, general knowledge) and estimated its complexity. Each model had an aptitude per topic, derived from public Arena Elo scores, and the router scored each model as:

$$\text{score} = \alpha \cdot \text{complexity} \cdot \sum_t p(t)\,\text{aptitude}_t \;-\; (1-\alpha) \cdot \text{normalised list price} \tag{B.1}$$

Here $p(t)$ is the topic probability and α sets the balance between quality and cost.

Complexity was first asked as a five-level question, but nearly all prompts scored between 0.3 and 0.55, so it barely separated easy from hard prompts. Asked instead as a yes/no question ("Is this task hard for an AI assistant to do well?"), with complexity as the probability of yes, $P(\text{hard})$, it separated hard prompts much better. Laya was then fine-tuned on 228 labelled prompts to improve the topic labels:

Table B.1. Classifier variants for the topic router.

Variant	Topic accuracy	Separation of hard prompts (AUC)	Time per prompt
Laya, five-level question	67%	0.77	134 ms
Laya, yes/no question	67%	0.92	133 ms
Fine-tuned Laya	76%	0.89	143 ms

Measured on 45 held-out prompts, of which only four were labelled hard, so the AUC (area under the ROC curve, from 0.5 for chance to 1 for perfect separation) values are rough.

Three other classifiers were tested on a common set of 368 prompts: NVIDIA's prompt task and complexity classifier (NVIDIA, n.d.), a BERT-embedding classifier and a local Qwen2.5-7B model (Qwen Team, 2024). They reached 63–75% topic accuracy and were not adopted. Because this test set differs from the 45 prompts used for Laya above, the two accuracy figures are not directly comparable.

Replayed on the Lönn routing benchmark, the topic router gave:

Table B.2. Topic router replayed on the Lönn routing benchmark.

Policy	Correct	Cost, 690 prompts (USD)
Topic router, Laya	78.1%	1.10
Topic router, fine-tuned Laya	78.4%	1.15
Topic router, given the true topic	78.6%	1.16
Always Claude Sonnet 5.5	88.1%	7.58
Oracle	94.1%	1.08

The router sent 94% of prompts to one model, DeepSeek V4.1 Flash, and even the true topic of every prompt improved it by only 0.5 points. **Lesson:** the routing logic, not the classifier, was the limitation. The complexity score varied too little to change the choice, four topics were too broad to separate tasks the models handle differently, and list prices predicted actual cost poorly. Each of these findings led to a design choice in the Lönn Router: task types, a routing table of measured accuracy and cost, and a value rule.

B.2 Task-type routing and the classifier

The first test of the new routing logic used the true task type of every prompt, to measure what the logic can achieve with perfect classification. Cross-fitted on the benchmark:

Table B.3. Task-type routing with the true task type, by λ.

Policy	Correct	Cost (USD)	Share of the most-chosen model
Topic router	78.1%	1.10	94%
Task-type routing, $\lambda = 80$	78.9%	0.40	41%
Task-type routing, $\lambda = 20$	84.1%	0.93	42%
Task-type routing, $\lambda = 10$	87.3%	2.88	35%

With the same models and the same answers, the new logic matched the topic router's quality at about a third of the cost, or gave six points more at lower cost, with choices spread across the pool.

The classifier was then chosen by comparison rather than up front, with a limit of 200 ms per prompt on a CPU. The candidates were trained on about 35,000 labelled prompts from unused parts of the public datasets, none of them in the Lönn routing benchmark. Multiple-choice and code questions were also included without answer options, so that the classifier learns the task rather than the format.

Table B.4. Classifier candidates for task-type routing.

Classifier	Task type correct	Router correct	Time per prompt
True task type (upper bound)	100%	84.1%	–
bge-small + linear layer (chosen)	**80.0%**	**84.0%**	**34 ms**
gte-modernbert-base + linear layer	84.7%	83.6%	up to 1.4 s
gte-modernbert-base + nearest neighbours	81.4%	83.5%	up to 1.4 s
Laya without training	31.1%	80.8%	260 ms

The smallest candidate brought the router within 0.1 points of the upper bound and was also the fastest. **Lesson:** once the routing logic was right, the classifier was no longer the bottleneck on the benchmark.

Everyday prompts look little like benchmark questions, and the classifier misread several of them, for example labelling "What is 17 * 23?" as instruction following. To measure this, 300 real user prompts from the WildChat dataset (Zhao et al., 2024) were labelled and set aside as a test set, and about 3,000 other WildChat prompts were added to the training data:

Table B.5. Classifier accuracy on 300 everyday WildChat prompts.

Training data	Task type correct (everyday test set)	Same model as with the true type
Benchmark-style prompts only	31%	55%
With everyday prompts added (chosen)	**51%**	**69%**

Lesson: benchmark results are optimistic for real traffic, and data from real use is what improves the classifier most.

B.3 Strategies tested against the gap to the oracle

With task types, the router reached 84% against the oracle's 94%. Of this gap, 68% lies in two task types, algorithmic programming and business writing, and most of it is between individual prompts within a task type, which a per-type table cannot capture. The strategies below were tested by cross-validated replay and kept only if they improved quality at equal cost.

Table B.6. Strategies tested against the gap to the oracle.

Strategy	Result	Kept?
Complexity from the benchmarks' own difficulty levels	No gain (84.5% at 1.19 USD vs 84.1% at 0.93 USD)	No
Difficulty known perfectly (upper bound)	At most +1 to +2 points	–
Complexity predicted from the prompt	Equal or 0.5–1 point worse	No
Two cheap models must agree, else a strong model answers	About +1 point, maths and multiple choice only	No
A cheap model judges writing before it is returned	Equal or up to 0.7 points worse	No
Check the answer, escalate at most once	Better at five of six budgets, 0.1 point lower at 1.00 USD; up to +1.7 points	**Yes**

- **Complexity.** Even perfect knowledge of each prompt's difficulty adds at most 1–2 points. A predicted difficulty splits the routing table into smaller cells, and the added noise cancels the gain. Complexity remains in the design but is switched off.
- **Agreement between two cheap models.** It helped only on maths and multiple choice, where answers can be compared, and the router already handled most of those prompts well.
- **A cheap writing judge.** It could detect weak texts, but no model was good enough to escalate them to: even the best one passes only 53% of the writing tasks.
- **Answer check.** Checking for answers that are cut off, empty or fail the prompt's own code examples was better at five of the six budgets and within 0.1 point at the sixth:

Table B.7. Router with and without the answer check, by budget.

Budget for 690 prompts (USD)	0.40	0.75	1.00	1.50	2.00	3.00
Router	79.1%	81.6%	83.9%	84.6%	85.3%	87.3%
Router + answer check	80.7%	82.3%	83.8%	85.7%	86.9%	89.0%

Lesson: most of the oracle's lead cannot be predicted from the prompt alone, since the oracle knows which model happened to be right. Checking the answer after the fact captures part of it, at the cost of a second call for a small share of requests.

B.4 Effect of λ on the Lönn Router

The table shows the Lönn Router, including the answer check, at different values of λ on the Lönn routing benchmark, cross-validated. A low λ favours quality and a high λ favours cost. The default is λ = 20.

Table B.8. The Lönn Router at different values of λ (with the answer check).

λ	Correct	Cost (USD)	Compared with always using Sonnet	Escalated
0	89.0%	6.90	+0.9 points, −9% cost	3%
10	**89.6%**	**3.91**	+1.5 points, −48% cost	6%
20 (default)	**85.8%**	**1.58**	−2.3 points, −79% cost	8%
30	84.0%	1.00	−4.2 points, −87% cost	9%
40	82.3%	0.82	−5.8 points, −89% cost	9%
50	81.4%	0.63	−6.7 points, −92% cost	9%
60	81.2%	0.53	−7.0 points, −93% cost	8%
70	81.0%	0.48	−7.1 points, −94% cost	7%
80	80.7%	0.44	−7.4 points, −94% cost	7%
90	80.6%	0.42	−7.5 points, −94% cost	7%
100	80.3%	0.39	−7.9 points, −95% cost	7%

At λ = 10 the router answers more prompts correctly than always using Sonnet, at about half the cost. Above λ = 50, raising λ lowers cost only slightly while quality keeps falling.

C Benchmark composition

Table C.1 lists the number of prompts drawn from each dataset and how they were sampled (see Section 3.4).

Table C.1. Benchmark composition: prompts drawn from each dataset and sampling method (fixed random seed 20261012).

Task family	Dataset	Prompts	Sampling method
Mathematics	MATH-500	150	Sampled by difficulty level: 20, 20, 30, 40 and 40 prompts from levels 1–5, respectively
Programming	HumanEval+	40	Random sample
Programming	MBPP+	40	Random sample
Programming	LiveCodeBench	120	Sampled by difficulty: 30 easy, 40 medium and 50 hard problems from the release-v6 additions
Knowledge	MMLU-Pro	120	Sampled approximately equally across 14 subjects, with 8–9 prompts per subject
Knowledge	GPQA Diamond	50	Random sample
Knowledge	ARC-Challenge	30	Random sample
Instruction following	IFEval	80	Random sample
Business writing	WritingBench	60	30 English prompts from Finance & Business and 30 from Advertising & Marketing
Total		**690**	

D Financial model

D.1 Summary

Table D.1. Financial model, summary sheet: inputs.

Item	Unit	Value	Notes in the workbook
Income			
AI spend with Lönn AI router	USD, 690 prompts	1.58	Final router with answer check, 690 benchmark prompts
AI spend with always Claude Sonnet 5.5	USD, 690 prompts	7.58	Always Claude Sonnet 5.5, same 690 prompts
Savings-based fee	% of customer savings	20.0%	Lönn AI keeps this share of the savings
USD/SEK exchange rate	SEK per USD	9.9566	Riksbank exchange-rate statistics, 30 September to 30 October 2026
Variable costs			
Card payment fee	% of total billed	1.5%	Estimate based on Stripe standard EEA card pricing; charged on the full bill including passed-through AI model cost
Card payment fixed fee	SEK per invoice	1.80	Estimate based on Stripe standard EEA card pricing; one invoice per customer per month
Currency conversion fee	% of AI model cost	2.0%	Estimate; SEK collected, USD paid to providers (Handelsbanken currency-exchange calculator)
Case customer			
Tokens per month	M tokens	500	“Middle of interval in report”
Input : output token ratio	input per output token	3 : 1	“Typical ratio 3:1” (LinkedIn article on token ratios in production AI)
Sonnet price, input tokens	USD / M tokens	2.00	Claude platform pricing page
Sonnet price, output tokens	USD / M tokens	10.00	Claude platform pricing page
Input tokens	M tokens	375	Tokens times 3/4
Output tokens	M tokens	125	Tokens times 1/4
AI spend at ceiling model	USD / month	2,000	Input tokens times input price plus output tokens times output price
Router cost ratio	share of benchmark spend	20.8%	Router cost divided by always-Sonnet cost on the 690 prompts (1.58 / 7.58)
Case customer AI spend with router (excl. service fee)	USD / month	**417**	AI spend at ceiling model times router cost ratio

Table D.2. Financial model, summary sheet: profit per 1 USD of routed spend (case customer).

Item	Unit	Value	Notes in the workbook
Income (savings-based fee)	USD / routed USD	0.759	From the income model: Lönn AI fee per USD of AI spend routed
Variable costs	USD / routed USD	0.047	From the variable cost model, at the case customer's spend
Contribution margin	USD / routed USD	**0.713**	Income minus variable costs, before fixed costs
Contribution margin %	% of income	93.8%	
Break-even volume	USD routed spend / month	592	Monthly routed AI spend (all customers) at which contribution covers fixed costs
Break-even customer volume	customers	**2**	Break-even volume divided by case-customer spend with router, rounded up

Table D.3. Financial model, summary sheet: profit by monthly routed AI spend (USD per month).

Routed AI spend (USD / month)	100	250	500	1,000	2,500	5,000	10,000	25,000	50,000	100,000
AI spend with always Claude Sonnet 5.5	480	1,199	2,399	4,797	11,994	23,987	47,975	119,937	239,873	479,747
Total billed to customer	176	440	880	1,759	4,399	8,797	17,595	43,987	87,975	175,949
Income (savings-based fee)	76	190	380	759	1,899	3,797	7,595	18,987	37,975	75,949
Variable costs	5	12	23	47	116	232	464	1,160	2,320	4,639
Contribution	71	178	356	713	1,783	3,565	7,131	17,827	35,655	71,310
Fixed costs	421	421	421	421	421	421	421	421	421	421
Profit per month	**−350**	**−243**	**−65**	**292**	**1,361**	**3,144**	**6,710**	**17,406**	**35,234**	**70,889**
Profit per routed USD	**−3.502**	**−0.973**	**−0.130**	**0.292**	**0.545**	**0.629**	**0.671**	**0.696**	**0.705**	**0.709**
One-time costs (USD)	241	241	241	241	241	241	241	241	241	241
Months to break even	n/a	n/a	n/a	1	1	1	1	1	1	1

D.2 Income model

The income model compares AI spend with the Lönn AI router against the benchmark (always Claude Sonnet 5.5) on the report's 690 test prompts, and calculates Lönn AI's savings-based fee.

Table D.4. Financial model, income sheet: the benchmark case (690 prompts).

Item	Unit	Value	Notes in the workbook
Step 1. What the customer spends with Lönn AI routing			
AI model cost with routing	USD	1.58	Passed straight through to the AI providers
Step 2. The benchmark: same prompts on the ceiling model			
AI spend at benchmark (Claude Sonnet 5.5)	USD	7.58	Same prompts on the ceiling model
Step 3. The savings, and Lönn AI's share			
Gross savings	USD	6.00	Step 2 minus Step 1
Lönn AI income (savings-based fee)	USD	**1.20**	Gross savings times fee %
Step 4. The customer's bill			
Total billed to customer	USD	2.78	AI model cost plus Lönn AI fee
Customer net savings	USD	4.80	What the customer keeps vs. the benchmark
Customer net savings	% of benchmark spend	**63.3%**	
Lönn AI income as % of routed spend	% of routed spend	75.9%	Lönn AI's fee relative to the AI model cost it routes

Table D.5. Financial model, income sheet: scaled to monthly spend at the same ratio as the benchmark (USD per month).

AI spend with Lönn AI router	100	250	500	1,000	2,500	5,000	10,000	25,000	50,000	100,000
AI spend at ceiling model	480	1,199	2,399	4,797	11,994	23,987	47,975	119,937	239,873	479,747
Gross savings	380	949	1,899	3,797	9,494	18,987	37,975	94,937	189,873	379,747
Lönn AI income	**76**	**190**	**380**	**759**	**1,899**	**3,797**	**7,595**	**18,987**	**37,975**	**75,949**
Total billed to customer	176	440	880	1,759	4,399	8,797	17,595	43,987	87,975	175,949
Customer net savings	304	759	1,519	3,038	7,595	15,190	30,380	75,949	151,899	303,797

D.3 Variable cost model

Variable costs are calculated per 1 USD billed to the customer, driven by the income model.

Table D.6. Financial model, variable cost sheet: variable cost per 1 USD billed (case customer, one month).

Item	Unit	Base	Rate	USD / billed USD	Notes in the workbook
Total billed to customer (incl. service fee)	USD / month	734			Case customer's routed spend times billed per routed USD
AI model cost share of the bill	% of billed	56.8%			Part of each billed USD passed through to AI providers
Card payment fee	% of billed	1	1.5%	0.0150	Charged on every billed USD
Card payment fixed fee	USD per invoice / bill	734	0.181	0.0002	1.80 SEK per invoice converted to USD, spread over the monthly bill
Currency conversion fee	% of AI model cost	56.8%	2.0%	0.0114	Only the AI model cost part is converted
Total variable cost per billed USD	USD / billed USD			**0.0266**	
Variable cost per 100 USD billed	USD / 100 billed USD			2.66	
Lönn AI income per billed USD	USD / billed USD			0.4317	Fee share of each billed USD (1 minus AI model cost share)
Variable costs as % of Lönn AI income	%			6.2%	

Table D.7. Financial model, variable cost sheet: variable cost per billed USD by customer AI spend.

AI spend with Lönn AI router	100	250	500	1,000	2,500	5,000	10,000	25,000	50,000	100,000
Total billed to customer (USD / month)	176	440	880	1,759	4,399	8,797	17,595	43,987	87,975	175,949
Card payment fee	0.0150	0.0150	0.0150	0.0150	0.0150	0.0150	0.0150	0.0150	0.0150	0.0150
Card payment fixed fee	0.0010	0.0004	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Currency conversion fee	0.0114	0.0114	0.0114	0.0114	0.0114	0.0114	0.0114	0.0114	0.0114	0.0114
Total variable cost per billed USD	**0.0274**	**0.0268**	**0.0266**	**0.0265**	**0.0264**	**0.0264**	**0.0264**	**0.0264**	**0.0264**	**0.0264**
Variable costs as % of Lönn AI income	6.3%	6.2%	6.2%	6.1%	6.1%	6.1%	6.1%	6.1%	6.1%	6.1%

D.4 Fixed cost model

Fixed costs do not depend on usage: monthly costs and one-time costs.

Table D.8. Financial model, fixed cost sheet: monthly and one-time costs in USD. SEK prices are converted at the USD/SEK rate in Table D.1.

Item	Unit	Price	Unit cost	Qty	Total	Notes in the workbook
Fixed costs per month						
Hosting (EU cloud)	per month	SEK 1,000	100.44	1	100.44	Higher-tier hosting plan, assumed sufficient for early volumes (eon.io blog on AWS hosting cost)
Claude Code Pro	seat / month	USD 20	20.00	6	120.00	Claude Pro, 20 USD per month billed monthly (includes Claude Code)
VS Code	seat / month	–	–	6	–	Free (Microsoft)
Company and admin costs	per month	SEK 2,000	200.87	1	200.87	Rounded up (driva-eget.se)
Total fixed costs per month					**421.31**	
One-time costs						
Development costs	one-time	–	–	1	–	Sunk cost; the router has already been developed
Registration fee (aktiebolag)	one-time	SEK 2,400	241.05	1	241.05	verksamt.se fee list
Total one-time costs					**241.05**	